universe, you might think, given our earlier discussion of the Coke and its carbon dioxide molecules, that the primordial gas was in a high-entropy, disordered state. But this turns out not to be true. Our earlier discussion of entropy completely ignored gravity, a sensible thing to do because gravity hardly plays a role in the behavior of the minimal amount of gas emerging from a bottle of Coke. And with that assumption, we found that uniformly dispersed gas has high entropy. But when gravity matters, the story is very different. Gravity is a universally attractive force; hence, if you have a large enough mass of gas, every region of gas will pull on every other and this will cause the gas to fragment into clumps, somewhat as surface tension causes water on a sheet of wax paper to fragment into droplets. When gravity matters, as it did in the high-density early universe, clumpiness—not uniformity—is the norm; it is the state toward which a gas tends to evolve, as illustrated in Figure 6.5.

Even though the clumps appear to be more ordered than the initially diffuse gas—much as a playroom with toys that are neatly grouped in trunks and bins is more ordered than one in which the toys are uniformly strewn around the floor—in calculating entropy you need to tally up the contributions from *all* sources. For the playroom, the entropy decrease in going from wildly strewn toys to their all being “clumped” in trunks and bins is more than compensated for by the entropy increase from the fat burned and heat generated by the parents who spent hours cleaning and arranging everything. Similarly, for the initially diffuse gas cloud, you find that the entropy decrease through the formation of orderly clumps is more than compensated by the heat generated as the gas compresses, and, ultimately, by the enormous amount of heat and light released when nuclear processes begin to take place.

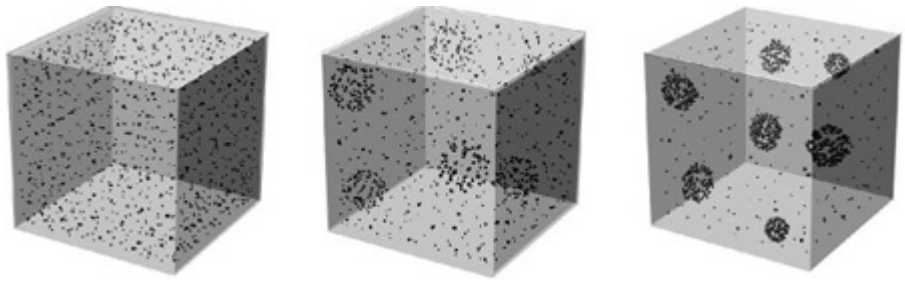


Figure 6.5 For huge volumes of gas, when gravity matters, atoms and molecules evolve from a smooth, evenly spread configuration, into one